4. Absolute pressure

Absolute pressure is the pressure measured with absolute zero taken as a reference. Hence, a perfect vacuum called absolute zero is the datum. It is equal to gauge pressure plus atmospheric pressure.

7.2.6 Pressure measuring devices

The atmospheric pressure is measured by using a simple mercury barometer.

The fluid pressure can be measured by using the following devices.

1. Piezometer
2. Manometer
   - a. Simple U-tube manometer.
   - b. Differential manometer.
   - c. Inverted differential manometer.
   - d. Sensitive manometer or Micro manometer.
3. Mechanical Pressure Gauges
   - a. Bourdon's tube pressure gauge.
   - b. Diaphragm pressure gauge.
   - c. Dead weight pressure gauge.

Activity 1

Prepare an album of different types of pressure measuring devices.

7.3 Flow of Fluids

7.3.1 Types of flow

When a fluid flows in a pipe line, it meets with resistance at the contact surface. The flow has different patterns of motion. The pattern of flow depends upon the conditions of flow and the surface conditions. The flow may be steady or unsteady, laminar or turbulent and uniform or non-uniform.

1. Steady flow

A steady flow is the one in which the velocity of fluid flowing per second through any section is constant.

2. Unsteady flow

An unsteady flow is the one in which the velocity of fluid flowing per second through any section is not constant.

3. Laminar flow

In this flow, each particle of fluid moves strictly in a straight line as shown in fig. It is also called as steam-line flow. This is possible when,

- The fluid is highly viscous or thick.
- The size of the conduit is small.
- The velocity of fluid is very low.



4. Turbulent flow

In this flow, the liquid particles do not have definite paths, but moves in a zig-zag way as shown in fig. Eddies or cross currents are created. Thus, the flow becomes very wavy along its path.



5. Uniform flow

In this flow, the magnitude and directions of velocities of fluid particles are same at all sections. Example: Constant discharge through a constant diameter pipe.

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